

Sustainable and Socially Resilient Minigrid Franchise Model for an Urban Informal Settlement in Kenya

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ABSTRACT

Kibera is a large informal settlement, in Nairobi, Kenya where electricity access is presently expensive, intermittent, and dirty. The context of Kibera also speaks to larger global dynamics of rapid urbanization, the creation of an urban poor, the transitory experience of informal settlements, and the role of non-governmental actors; each of which provides challenges to traditional sustainable urban planning. This research explores the opportunity for environmentally sustainable and socially resilient development in Kibera through a mixture of simulation and field-based assessment of a hybrid minigrid for a community empowerment facility—the Kibera Town Centre (KTC). Following the installation of one of the area's first grid-tied solar-battery systems in 2018 and a national policy change in 2019, the future operation and potential expansion of KTC's energy system affords an opportunity to examine the feasibility of a 'franchise minigrid' model to power community loads including schools and small businesses. Using the HOMER Pro minigrid simulation tool, we find that the levelized cost of energy will decrease by 60% with optimal minigrid expansion, thereby achieving costs less expensive than the prevailing utility rates. We then propose a novel, community-empowering, and profitable 'minigrid franchise' business model that can provide supplemental income and employment to neighboring businesses and homes. This research uniquely combines primary data, simulation, policy opportunity analysis, and business models in collaboration with the community to contribute to literature enabling electricity access for informal settlements and informing the context of sustainable and resilient development for the urban poor.

Keywords: Sustainable cities, renewable energy, electricity access, informal settlements, equitable employment, community minigrids, sustainable development

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1. INTRODUCTION

In 2015, the United Nations adopted the goal to “ensure access to affordable, reliable, sustainable energy for all” as part of its sustainable development goals (SDGs) for 2030 (UN General Assembly 2015). While approximately 840 million people lack any access to electric-

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ity globally (World Bank 2019), several billions remain in situations where their electricity is neither affordable, nor reliable, nor sustainable. One particular subset, that of the millions of individuals in informal settlements in the cities of developing countries, live in challenging circumstances that are not easily solved by traditional sustainable urban development programs. Additionally, rapid population and demand growth will amplify the harmful effects of climate change, which in the present and future disproportionately affects the world's poorest communities (Casillas & Kammen 2010; Roberts 2009). In order to fully achieve sustainable cities, we must account for the enduring presence of informal settlements, which need thoughtful approaches and directed solutions.

Approximately 24% of the global urban population live in informal settlements, also referred to as slums¹ (UN Stats 2020). Informal settlements are generally defined by their: impoverished populations; severe food, shelter, and employment insecurity (Uwizeye et al. 2020); high population density; lack of formal infrastructure or services; and continuously transitory nature. The diverse populations of informal settlements can include refugees as well as resident populations trapped in cycles of poverty. At the end of 2019, approximately eighty million individuals were forcibly displaced worldwide as a result of persecution, conflict, violence or human rights violations (UNHCR 2020). In Kenya, the rural-to-urban wage differential, opportunity-driven migration, regional conflict, and intra-urban displacement has led to the creation of large, enduring slums where 46.5% of Kenya's urban population resides (UN Stats 2020). Kenyan government records indicate that the 2007 political crisis internally displaced 666,000 people, 47% of which are considered 'integrated', often in informal settlements such as in Kibera. Outside of camps, internally displaced people are left without access to assistance programs that their status would otherwise warrant, while facing discrimination and insufficient access to basic services (Kamungi 2013).

These large, enduring, informal settlements present numerous challenges to sustainable urban development. Because they remain outside traditional structures of governance and ownership, many existing policies and programs are often infeasible to implement. Other characteristics of informal settlements, such as the transitory nature of the population and informality, mean that inflexible solutions quickly become inappropriate and subsequently abandoned. The process of designing resilient and sustainable urban solutions must account for the different needs, requirements, and barriers common to informal settlements.

This article provides a case study of an environmentally sustainable and socially resilient minigrid distribution model to serve communities in an urban informal settlement in Kibera, Nairobi, Kenya. We integrate a wide range of methodological and technical approaches including surveys, sensor-based monitoring, energy system simulation, policy analysis, and evaluation of a proposed business model. Our primary data collection and analysis provides key insights into our understanding of informal settlements which suffers from a severe lack of data (Butera et al. 2016). Collected in collaboration with the Kibera Town Center (KTC), we use unique primary electricity production and usage data to analyze informal settlement electricity demands. However, unlike previous literature in informal settlements that focuses on residential energy, this research focuses on critical, centralized community-scale energy that supports schools, water delivery, and livelihoods. Furthermore, under a policy change in Kenya's Energy Act 2019, this minigrid now has the opportunity to expand and sell power to the local community under new business models. When scaled, our proposed innovative minigrid

1. See Mitra et al. (2017) and Montoya et al. (2020) for more discussion around the terms 'slum' and 'informal settlement'. Mitra et al. focuses on Kibera, Nairobi, Kenya, while Montoya et al. focuses on Bogota, Colombia.

franchise business model can support microenterprises and return agency to slum dwellers (Cubitt 2013).

Our research is guided by the following questions:

- *What is the existing electricity landscape of the community center, local schools, and local shops?*
- *What is the optimal amount of solar photovoltaic (PV) capacity to serve KTC and the adjacent Olympic Primary and Secondary schools?*
- *Which technical minigrid scenarios provide the lowest cost of electricity? Is it at a higher or lower cost than grid extension?*
- *How can a community minigrid franchise model—made possible through Kenya’s Energy Act 2019 and approached in collaboration with communities, non-governmental organizations (NGOs), and government stakeholders—showcase environmentally sustainable and socially resilient planning for urban informal settlements?*
- *How can results from this case study inform other resilient and sustainable infrastructure for informal settlements?*

We find that the community center, local schools, and individual shops surrounding the center respectively consume an average of 324 kWh, 274 kWh, and 2 kWh each daily. Through minigrid simulation and analysis using HOMER Pro, we find that the levelized cost of energy (LCOE) will decrease by 60% with optimal minigrid expansion, thereby demonstrating the economic feasibility of expansion. Through stakeholder mapping efforts and a framework analysis, we explore how a community franchise model can additionally showcase environmentally sustainable and socially resilient planning for informal settlements.

The remainder of this article is divided into seven sections: Section 2 reviews relevant literature on sustainable electricity for urban informal settlements, minigrid simulation for electricity access, and energy-related franchise models; Section 3 provides case study background; Section 4 describes the minigrid simulation and analysis methods; Section 5 describes the analysis results; Section 6 introduces the minigrid franchise model and examines its feasibility; Section 7 notes limitations and opportunities for future work; and finally, Section 8 concludes this article.

✎ 2. LITERATURE REVIEW ✎

Literature on sustainable cities often emphasizes green building design and construction (Bardhan et al. 2018), clean mobility (Melkonyan et al. 2020), clean cooking (Osano et al. 2020), the urban heat island effect (Li et al. 2020; Litardo et al. 2020), and more recently, smart city management techniques (Peprah et al. 2019; Madlener & Sunak 2011). While broad, the conventional literature generally excludes the nuances of informal economies common to many of these same cities, particularly in the developing world. Urbanization that outpaces city planning gives way to the proliferation of the informal economy and illegal building sector (Shahraki et al. 2020). This negatively impacts existing city-owned infrastructure that serves electricity and water needs by increasing the risk of damage and theft (Madlener & Sunak 2011). With increased urbanization in developing countries, efficiently and equitably supplying the growing electricity demand remains a challenge (Lau et al. 2017).

Research that focuses on informal settlements offers unique insights for strategic urban planning, urbanization, and energy planning (Simwanda & Murayama 2018; Madlener &

Sunak 2011). Not only are these economically disadvantaged communities often living in precarious housing infrastructure, they also often face political disadvantages from internal displacement status and lack of citizenship. Therefore, in the context of energy poverty and political power, informal settlements parallel that of refugee camps (Huq and Miraftab 2020; Bird et al. 2017; Mimmi 2014). These barriers can undermine the effects of increased urban informal economic activity on the quality of their built environment (Farinmade et al. 2018). Furthermore, people living in informal settlements often have a distrustful, complicated relationship with government services, especially when state infrastructure projects often forcibly remove slum-dwellers, destroying livelihoods and senses of security (Mitra et al. 2017; Munguti & McGranahan 2002).

The varied landscape of energy provision in informal settlements includes multi-dimensional challenges ranging from public health concerns, gender inequalities, disproportionately subpar service, to limited energy choices (Bird et al. 2017). Energy poverty is a well-documented source of public health insults such as poor indoor air pollution, lack of sanitation facilities, time-consuming fuelwood collection, and the lack of refrigeration for food and medical needs—all of which exacerbate gender inequalities (Ezzati & Kammen 2001; Saini et al. 2019; Sovacool 2012). Recent studies on electricity reliability reveal the extent to which electricity service is differentiated across economic classes. Aidoo and Briggs (2018) found that the richest neighborhoods in Accra, Ghana received ten more hours of electricity per day than the poorest (which were informal settlements). Similar results were found in Nairobi, Kenya by Ferrall et al. (2020). These large differences were attributed to issues stemming from informal connections, unequal distribution system maintenance schedules, and the potential for powerful decision makers to target the effects of load shedding and rolling blackouts to “*less-influential*” customers (Bird et al. 2017; Butera et al. 2016). This lack of reliable access to electricity limits economic development and access to basic services (Gunning 2014; Moyo 2013).

Traditional urban planning has allowed the centralized electricity grid to dominate in urban areas, while minigrids have become the cost-competitive solution for rural contexts over the past decade of energy access planning (Mandelli et al. 2016). Though centralized energy systems were historically economically favorable, the continued falling cost of decentralized energy systems necessitates a deeper examination of the co-benefits and opportunities from sustainable *urban* minigrids. Decentralized renewable energy systems, such as solar-home-systems, are also becoming an increasingly popular pathway for providing lighting services (Alstone et al. 2015), however, the systems available to those living in informal settlements are often too expensive or have lifetimes and power capacities that are too small to support businesses.

Community minigrids powered by renewable energy are being increasingly deployed in areas without reliable electricity in Sub-Saharan Africa (Mandelli et al. 2016). The HOMER Pro energy simulation software, briefly described in Section 4.1, is a well respected tool for minigrid researchers and developers (Liu et al. 2018; Singh et al. 2018). When used to analyze systems in developing countries, the target is often for rural electrification and off-grid systems, but rarely for informal settlements or displaced populations (Mandelli et al. 2016; Mimmi 2014). Many studies identify the lack of data on displaced populations as a significant challenge to adequately addressing their energy needs with infrastructure projects (Butera et al. 2016; Liu et al. 2018; Salehin et al. 2011; Singh et al. 2018; Moner-Girona et al. 2018). As a solution to maintenance, data collection, and data sharing challenges for existing on- and off-grid infrastructure, some studies introduce advanced metering infrastructure, system flexi-

bility through voltage control, and demand side management to compensate for non-technical losses, illegal connections, and uncollected debt. However, such solutions may not be financially accessible to the urban poor (Lee et al. 2019; Saini et al. 2019; Welsch et al. 2013).

For these reasons among many, sustainable electricity planning for urban informal settlements necessitates thoughtfully designed solutions. In particular, the creation of appropriate business models and implementation plans that are affordable and meaningful for these communities requires market analysis and socially-conscious design (Clements et al. 2019). Most literature on business models in these contexts focuses on product franchise models, such as Pay-as-you-go financing schemes for electricity access, which have been successful in deploying small solar lanterns, solar-home-systems, and rural minigrids in India and East Africa (Alstone et al. 2015; Yadav et al. 2019). Resilient networks of peer-to-peer energy trading in these communities are anticipated to give agency to slum dwellers and decrease utility theft. However, while these business models are emerging, peer-to-peer models have yet to be proven (Cubitt 2013).

In summary, the majority of the relevant literature focuses on broader sustainable urban planning such as clean mobility and green building design; estimating electricity demand (Boait et al. 2015); designing rural stand-alone minigrids; and conceptualizing product franchise models. We find a significant gap in the literature regarding technically supported business analyses that appropriately examine sustainable urban minigrid potential for informal settlements. The remainder of this article demonstrates the opportunity for sustainable, resilient, urban minigrids to provide affordable, reliable electricity for informal settlements in ways that are more technically, economically, and socially appropriate than traditional urban development solutions.

✦ 3. RESEARCH LOCATION CONTEXT—KIBERA, NAIROBI, KENYA ✦

We evaluate the potential for renewable energy to meet local energy demands in the Olympic neighborhood of Kibera, Kenya through the expansion of an existing community-facing program—KTC. KTC is the pilot of the Human Needs Project,² an international non-profit organization that focuses on providing basic services (clean water, sanitation, energy) and empowerment services (business skills training, wifi, green marketplace, money lending) in a self-sufficient, replicable manner. Approximately 600 residents from across Kibera rely on the services that KTC provides each day. Strong partnerships with enduring infrastructure like KTC provides an opportunity for resilient electricity planning to serve characteristically transient populations.

Through a collaboration between UC Berkeley and KTC spanning several years, this research has ensured the participation and engagement of the local community. Unique primary data generated through our partnership with KTC informed this research; specifically: residential, community, and business energy supply norms in Kibera; school and business energy audits, electricity outage history, utility bills, generator fuel costs, daily load profiles, and solar and battery installation costs. These unique datasets provide a novel contribution to a space with little documentation or empirical evidence and can be found in supplemental information (SI) to support future research. We outline the extensive monitoring and evaluation that

2. Human Needs Project <http://humanneedsproject.org/>.

is integral for maximum impact and justification for the system's solar capacity expansion into Kibera, which can serve as a model for similarly impacted populations.

3.1 Status-quo electricity service for Kibera's residents

Similar to most informal settlements, community members in Kibera often rely on community electricians to informally (illegally) tap the utility's power lines for their electricity. These informal connections serve as significant burdens for utilities, but are easily obtained by individuals. Mimmi (2014) identified the major barriers that prevent households from obtaining a standard utility connection in a similar context of an urban slum in Mumbai.³ The barriers range from affordability, to local social and political pressure, to the house's ownership status and precarious construction. However, the live wires and poor infrastructure conditions associated with unmonitored electrical connections significantly increase safety risks.

The community electricians setting up informal connections in Kibera have historically charged households a flat monthly rate significantly less than the utility's fees (Mimmi 2014). The electricity tariff for grid electricity in Kibera from Kenya Power & Lighting Co. (KPLC)⁴ has peak and off-peak rates, an energy charge, and fluctuating monthly surcharges.⁵ The Global Partnership on Output-Based Aid and a 2015 World Bank IDA grant to KPLC lowered the upfront cost of an electricity connection in Kibera from \$150 to \$12 USD (1,165 KSH) (Redfern 2015). However, overall, utility electricity can still be three times more expensive than the rates offered by community electricians.

In addition, Kibera is subject to notoriously poor grid electricity reliability. While an average customer in Kenya can expect 80.9 hours of power outages/year,⁶ an average customer in Kibera can expect 630.5 hours/year.⁷ The closest grid sensors to KTC report 377.3 hours of power outages/year. These are equivalent to 4.3%–7.2% of the time in an outage state, with the average duration of outages ranging between 1.6–1.8 hours (Ferrall et al. 2020).

Finally, the majority of the energy requirements for homes in informal settlements go towards fuel for cooking and electricity for lighting. The low levels of electricity demand make it difficult for residents to pay back the upfront cost of individual decentralized renewable energy systems as alternatives to the unreliable grid electricity (Bird et al. 2017).

3.2 Schools in Kibera

Schools tend to be in central locations in Kibera, with higher electricity demands and with roof space ideal for community solar installations. The Kenyan government provides stipends to government schools to pay for expenses like electricity, utilities, and salaries. There are approximately 359 schools in Kibera, only 14 of which are government owned (Schools for Everyone). The remaining schools are owned and run informally. Through energy audits, we

3. 32% stated the cost of informal connections was less, 29% stated the cost of legal connections was too high, 14% stated their connection was provided by their landlord, 8.9% stated they had failed at earlier attempts to get a legal connection, 7.2% stated the household lacks a proper address, and 6.4% stated they could not get a legal connection because their household structure was illegal.

4. Kenya Power and Lighting Company (KPLC) - <https://kplc.co.ke/>.

5. Available in SI, Section 3, Tables 5.A, 5.B, and 5.C.

6. Reported by KPLC to the World Bank Doing Business Surveys, See Ferrall et al. (2020) for more information.

7. Reported by the Electricity Supply Monitoring Initiative (ESMI) Kenya Initiative <http://esmikenya.org/>, See Ferrall et al. (2020) for more information.

found that the 14 government schools combined require –1.4 MWh/day and have roof space for installing over 1 MW of solar capacity.

FIGURE 1
Solar on Olympic Primary School.



Proposed design of 69 kW of solar panels (each rated 320 W) on the roof of the Olympic Primary School Buildings 1, 2, 3, and 7. There are 17 buildings in the school complex, with over 2500 m² of roof space, allowing for up to 600 kW in installed solar capacity.

The Olympic Primary and Secondary Schools neighboring KTC serve roughly 4,000 students and together demand 274 kWh/day. The schools’ operating costs are dominated by electricity fees—money which could be better spent on education and teacher salaries. Figure 1 displays an aerial view of Olympic Primary School with overlaid solar panels to estimate the solar capacity.

3.3 Businesses and community in Kibera

Energy audits of the 50 shops surrounding the Olympic Primary School indicate that each consumes approximately 2 kWh electricity daily. As shown in Table 1, 90% of the shops pay community electricians a flat rate of \$3–4 USD monthly⁸ to informally connect to the KPLC distribution lines, resulting in paid community members but at a higher risk of fire and death from live wires.

Unlike residential electricity demand, the small commercial sector has higher electricity loads. Oftentimes livelihoods are directly dependent on specific appliances and service quality in their stores (Booth et al. 2018). Common electric appliances for the small commercial sector include hair dryers, electric shavers, and refrigeration (See SI). With the expansion of the KTC

8. Conversation with Kibera resident, 6/20/2018.

TABLE 1

Percentage of Kibera businesses using different types of electricity connections.

Electricity connection types	Percentage (%) of surveyed Kibera businesses
Actively using KPLC Meter	10 %
Installed but inactive KPLC Meter	26 %
Unregistered electricity connection	90 %

Survey results show the distribution of electricity connection types. Connection rates per month (\$USD/month) range from \$3.5/month for unregistered connections to \$9.09/month for registered KPLC connections that use an average of 1.6 kWh/day.

minigrid, the fifty shops surrounding KTC could have their productive electricity demands met in an inclusive, community minigrid that incorporates the users' ability and willingness to pay as well as other social and economic considerations. Additionally, the aggregation of many smaller energy demands across a community results in a more predictable total load, which decreases investment risk and allows for appropriate, affordable financing options for the minigrid overall.

3.4 Future of KTC's energy model under new policy opportunities

KTC's recent efforts show promising results for the expansion of, and employment in, the renewable energy sector. In October 2018, KTC employees finished installing a 35 kW hybrid solar-battery system to increase energy independence, decrease the cost of energy, improve reliability, and promote renewable energy, education, and employment in the community. Prior to the installation in 2017, KTC spent a significant portion of its revenue on electricity—more than \$23,000 USD annually. 10% of the expenditure went to the diesel generator, which was manually turned on whenever KPLC services were inadequate (See SI). Since the installation, KTC's energy costs have decreased by 45% (\$10,000 USD annually) and utility energy consumption (in kWh) has decreased by 50%. However, from January-August 2019, they still paid a total of \$2,600 USD in generator fuel costs.

Kenya's recent Energy Act 2019⁹ introduced a key policy change allowing businesses to legally generate and sell electricity directly to customers, bypassing the utility notorious for charging high rates and exacerbating energy justice issues. With the recent legislative backing for net metering and a feed-in-tariff, KTC could profitably expand their existing solar and battery system to the nearby school and local businesses. By doing so, they can increase their community's energy independence and provide renewable, reliable, and affordable electricity while prioritizing formalized local employment and education opportunities. Through the opportunity provided by this policy direction, we investigate the system expansion through HOMER Pro simulation and analysis and then propose an innovative minigrid franchise business model that can sell electricity to the surrounding communities.

4. MINIGRID SIMULATION AND ANALYSIS METHODS

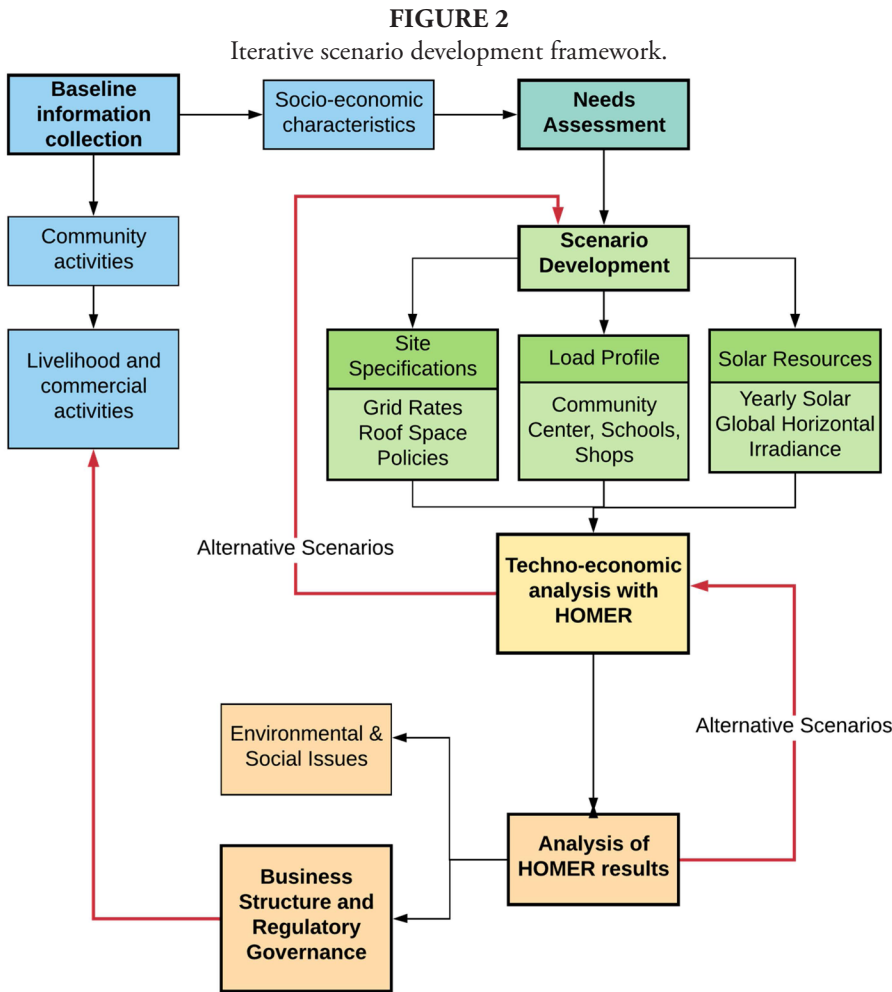
The following section outlines our minigrid simulation and analysis methods and provides a baseline for energy costs and thereby justify our subsequent contribution of an environmentally sustainable and socially resilient minigrid franchise model. However, even in this seemingly quantitative analysis, we use a mixed-methods approach.

9. Excerpts of Kenya's Energy Act 2019 can be found in Appendix B.

After a problem identification exercise with stakeholders, we integrate their perspectives to develop realistic and policy-relevant scenarios using the iterative framework in Figure 2. We then use a well-respected minigrid simulation and optimization tool to evaluate the minigrid expansion’s techno-economic viability employing extensive datasets collected with community collaboration (See SI). We validate the simulations against empirical data on the KTC system. Then, our techno-economic analysis of KTC’s energy usage predicts monthly utility energy purchases and the minigrid’s ability to meet demand with the 35 kW solar-battery system. A sensitivity analysis further evaluates the impact of solar resource variability on monthly grid purchases and energy supply. Finally, we analyze the scenario results to suggest a novel financial model that focuses on the community’s sustainable development needs.

4.1 Scenario Development

Through Figure 2’s iterative framework, we arrived at the five scenarios in Table 2. Beginning with ‘Baseline Information Collection’, we gathered qualitative and quantitative site-specific information. These informed the ‘Needs Assessment’ for the proposed minigrid simula-



Informed by Bhattacharyya (2015).

TABLE 2
Scenarios. Descriptions of the five developed scenarios and their acronyms.

Scenario	Description	
Reference	Ref	Only KTC loads served, net metering applied
Optimal Solar Expansion	OptSol	KTC + school loads served, optimal amount of additional solar capacity
Load Increase	LoadInc	KTC + school loads served, allowing no expansion of current system
Optimal Solar Expansion Renewable Only	OptSolRE	KTC + school loads served, optimal amount of additional solar and battery capacity, constraint of 100% renewable energy
Community Minigrid Conservative	ComCons	KTC, schools, + 50 shop loads served, optimal amount of additional solar and battery capacity

tion. Next, we developed scenarios based on the identified needs and quantitative information such as solar resource, demand, and grid rates. Depending on the constraints, the optimized simulation may not have a feasible solution or give interpretable results, necessitating the iterative nature of scenario development as represented by the red arrows. Once established, additional scenarios built from the ‘status quo’ of the KTC system. For example, in one scenario, minimum solar capacity is set to the existing 35 kW and the simulation is optimized for the maximum. The iterative framework narrowed our analysis from an infinite set of possibilities to five realistic scenarios. Finally, we evaluated potential business models using the simulation results such as by evaluating sensitivities to the sell-back electricity rate.

4.2 HOMER Pro simulation tool

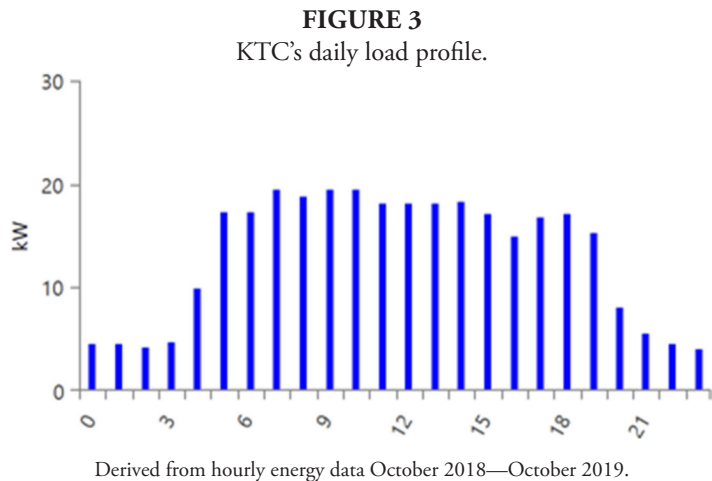
Our simulation of the expansion of the KTC minigrid uses HOMER Pro,¹⁰ a global standard for optimization of solar-storage systems, minigrids, and distributed energy projects that identifies the least-cost combination of multiple energy sources to reliably meet specified electric loads (Lambert et al. 2006). After intensive data input, HOMER Pro chronologically simulates hourly time steps for an entire year, optimizing for LCOE as well as installation and operational decisions. By combining engineering and economics, this tool assists in evaluating realistic scenarios for sustainable distributed energy systems. The HOMER Pro simulation tool is particularly well suited to verify scenario costs for this research because it is globally respected, has been rigorously designed and tested, and can be easily modified for future minigrid development in other urban informal settlements.

4.3 Additional simulation inputs and constraints

In addition to globally available information such as solar irradiation, we collected unique datasets in partnership with KTC covering small businesses, load profiles, and electricity service quality. Our primary datasets include documented utility outage history, energy audits from the community, solar production, battery usage, equipment costs, generator receipts, and utility bills. The equipment costs used in the minigrid system simulations come from KTC’s 2018 installation of their 35 kW grid-tied solar-battery system and are therefore locally specific and contextually realistic. These costs are considered conservative estimates of future costs as solar and storage prices decrease globally (Moner-Girona et al. 2006). Full documentation available upon request.

10. HOMER Pro. <https://www.homerenergy.com/products/pro/index.html>.

One week of hourly data showed an average demand of 220 kWh/day with a 25 kW peak. As more electricity data was collected, estimates were updated. Records from June 2018 to August 2019 show an average daily demand of 320 kWh. Figure 3 shows a core simulation input—the daily load profile of KTC generated from data collected over one year.



Device-level energy audits and utility bills showed that schools demand between 100 and 200 kWh/day of electricity. Similarly scaled energy audits and a qualitative survey of 50 shops in Kibera revealed that small-business owners in the community demand an average of 2 kWh/day (See SI). Table 3 details the average and peak demand for the different loads in the simulation.

TABLE 3
Average and peak energy demands.

Load	Average Energy Demand (kWh/day)	Peak Power Demand (kW)
KTC	324 kWh/day	36 kW
Secondary School	147 kWh/day	21 kW
Primary School	127 kWh/day	18 kW
50 shops (aggregated)	100 kWh/day	16 kW
Total Community	698 kWh/day	79 kW

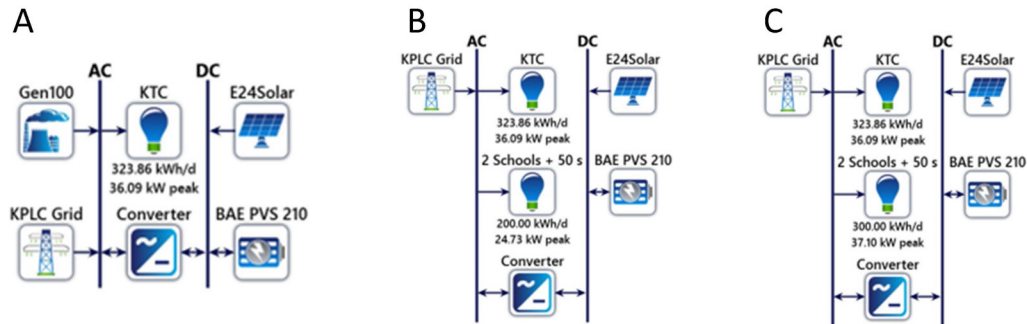
Data was consolidated from sensors, utility receipts, and extensive energy audits to determine loads for KTC, the schools, and the aggregated shops.

Additional constraints stem from the Energy Act 2019 which limits electricity producers to 1 MW. The roof space on Olympic Primary and Secondary School is estimated to be able to accommodate up to 600 kW in installed solar capacity (See Figure 1). Each shop can install roughly 600 W of conventional rooftop solar panels (See SI).

By legalizing net metering and licenses for commercial entities to sell electricity, the Energy Act 2019 opens the door for innovation minigrid designs. We use this key policy change to consider the potential for environmentally sustainable and socially resilient business models that leverage existing infrastructure and enduring community institutions to produce local jobs and sustainable energy to vulnerable and transient populations. Electricity rates under

this new policy are an essential to the simulations. KTC’s electricity rate from KPLC is consistent with other “small commercial” businesses. However, the base rate used in the simulation (\$0.156 USD/kWh) is conservative since it does not include the additional taxes and levies that fluctuate monthly.

FIGURE 4
Simulation representations for select scenarios.



A. *Ref*, showing optimal operation of the existing KTC system, B. *OptSol*, serving KTC and school loads with optimal solar capacity, and C. *ComCons*, serving KTC, school, and microenterprise loads with optimal solar and batteries.

Finally, we simulate 12 V, 2.64 kWh nominal capacity lead acid batteries emulating KTC’s installed system. Grid reliability inputs for the simulation are informed by nearby sensor data from ESMI and corroborated with three years of manually recorded outage history at KTC.

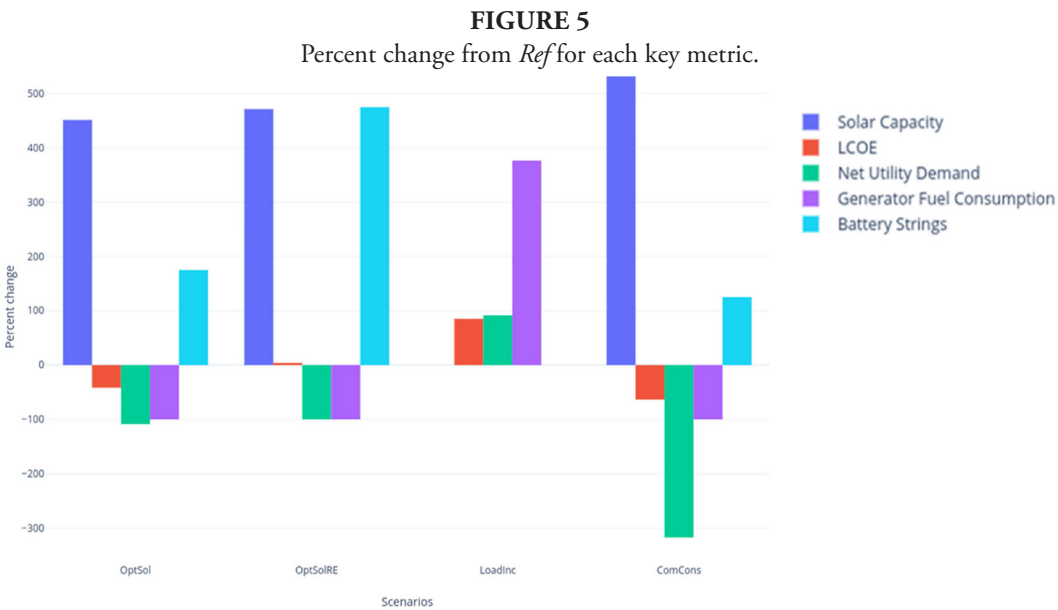
5. MINIGRID ANALYSIS RESULTS AND DISCUSSION

Table 4 summarizes results from the least-cost optimization for the five scenarios in Table 2, and Figure 5 further visualizes the percent change from the reference scenario. All simulations have zero capacity shortage, meaning that all electricity demand is served. The results reveal that expanding the 35-kW hybrid solar-battery system is economically and technically feasible, and will thereby alleviate energy costs, create energy independence, increase electricity reliability, and decrease emissions. Full simulation outputs available upon request.

TABLE 4
Optimized results for each scenario.

Scenario	Solar Capacity	LCOE (USD/kWh)	Net Utility Demand	Renewable output (% generation)	Generator fuel consumption	Battery size (strings)
Ref	35 kW	\$0.199 /kWh	35,313 kWh/yr	58.1 %	6,151 L	8
OptSol	193 kW	\$0.116 /kWh	−3,018 kWh/yr	87.3 %	0 L	22
OptSolRE	200 kW	\$0.207 /kWh	0 kWh/yr	100 %	0 L	46
LoadInc	35 kW	\$0.368 /kWh	67,576 kWh/yr	25.1 %	29,317 L	8
ComCons	221 kW	\$0.073 /kWh	−76,602 kWh/yr	91.4 %	0 L	18

Scenarios detailed in Table 2. Solar capacity, LCOE, net utility demand, renewable output, generator fuel consumption, and battery size highlighted based on original research questions. Figure 5 shows the percent change of each additional scenario from *Ref*.



Solar Capacity (dark blue) increases in every scenario. Changes to LCOE (red) vary but are generally small. Generator fuel (purple) increases only in *LoadInc*. Net utility demand (green) is mostly negative, meaning the system sells more than it pays to the utility, especially in *ComCons*. More storage (light blue) needed particularly in *OptSolRE*.

The least-cost way to achieve the Reference scenario (*Ref*—meeting the current energy needs of the KTC using its current infrastructure) had 60% of the generation served by solar and the remainder by the utility and generator. The LCOE was \$0.199 USD/kWh. Total actual generator usage is consistent with the results of the simulation, indicating that KTC is already using their system optimally. However, maintenance outages are not represented in the simulation. In reality, the generator is used when the rest of the minigrid system needs maintenance in addition to reacting to inadequacies in the utility resources, solar, and batteries.

When the school loads are added and installed solar is optimized in the Optimal Solar Expansion (*OptSol*) scenario, the total solar generating capacity reaches 193 kW with no usage of the generator, at an LCOE of \$0.116 USD/kWh. This indicates that the system will economically benefit from being expanded, avoiding the expensive fuel costs of the diesel generator. *OptSol* leaves 58% of Olympic Primary School’s roof space and 100% of Olympic Secondary School’s roof space available, providing for future expansion.

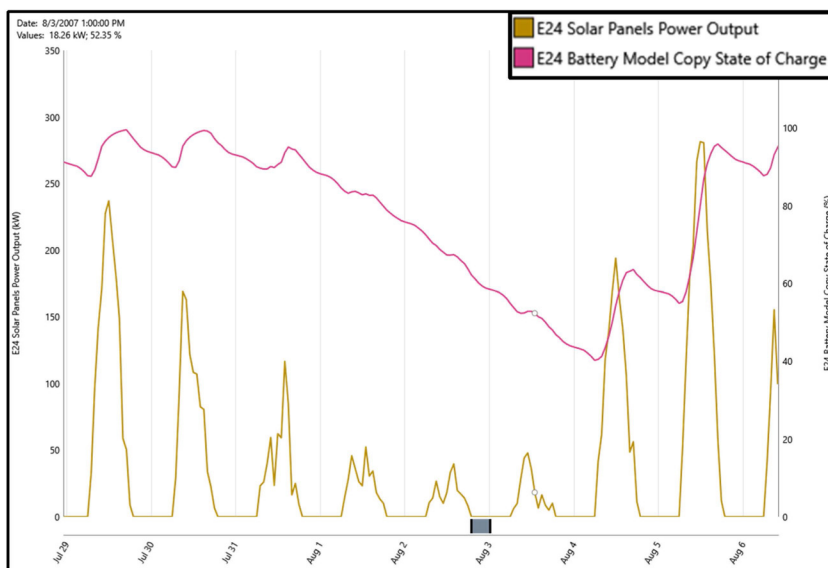
The Optimal Solar Expansion Renewable Only (*OptSolRE*) scenario simulation was constrained by 100% renewable energy serving both the school and KTC loads. The optimization installed 200 kW solar with significant storage (46 battery strings). LCOE is highly dependent on the addition of storage, therefore *OptSolRE* has the highest LCOE of all scenarios evaluated. However, since this scenario eliminates carbon production during energy generation, there are unquantified but positive local and global benefits for environmental and human health. For comparison, if the load of the current system is increased to include the schools without allowing changes to system capacity, i.e. the Load Increase (*LoadInc*) scenario, we find there is significantly higher generator usage and LCOE than in *Ref* because of the poor utility reliability.

The Conservative Community Minigrid (*ComCons*) scenario, which serves KTC, school, and business loads with optimal installed solar and batteries, achieved the lowest LCOE. The LCOE decreased from \$0.199 USD/kWh to \$0.073 USD/kWh when load and generation

were added to *Ref*, supporting other findings that larger solar PV installations in Africa decrease electricity's unit cost (Moner-Girona et al. 2006). Capitalizing on the opportunity made possible by the Energy Act 2019, this simulation injected 76,602 kWh/yr of excess solar into the grid, aiding Kenya's overall renewable energy transition (Carvalho et al. 2017), and allowing the potential for future growth in the minigrid franchise model or the ability to serve other community loads. Net present value (NPV) calculations detailed in Appendix C indicate that building *ComCons*'s system from no existing infrastructure would be \$31,000 USD more expensive than building *Ref*'s system from scratch. However, this difference is less than expected community benefits over the system's 25-year lifespan. A better estimate of the true economic value of *ComCons* evaluates the NPV of moving-forward, rather than sunk, costs. When NPV is calculated from the perspective of analyzing only the expansion of *Ref* into *ComCons* as compared to the going-forward costs of *Ref*, KTC could save over \$38,000 USD over the next 25 years by implementing the community minigrid instead of maintaining the status-quo.

Though temperate, Nairobi has rainy months (April, May, November, and December) that can significantly affect the system's ability to meet demand. The sensitivity analysis on the impact of solar resource variability found that the lowest state of charge of the battery bank for the *ComCons* simulation was 52.35% after a series of 3–4 days with little solar power output due to rainy and cloudy weather in July and August. Therefore, we find that the optimized system is sufficient to meet loads even during critically low-solar-resource periods. Figure 6 details a typical simulation for a series of cloudy days.

FIGURE 6
Simulation of cloudy days.



Solar panel power output (orange) and battery state of charge (pink) on a critically cloudy, 4-day period demonstrates system resilience. Solar panel output calculated from solar radiation data at exact longitude and latitude.*

*HOMERenergy Solar GHI Resource www.homerenergy.com/products/pro/docs/latest/solar_ghi_resource.html.

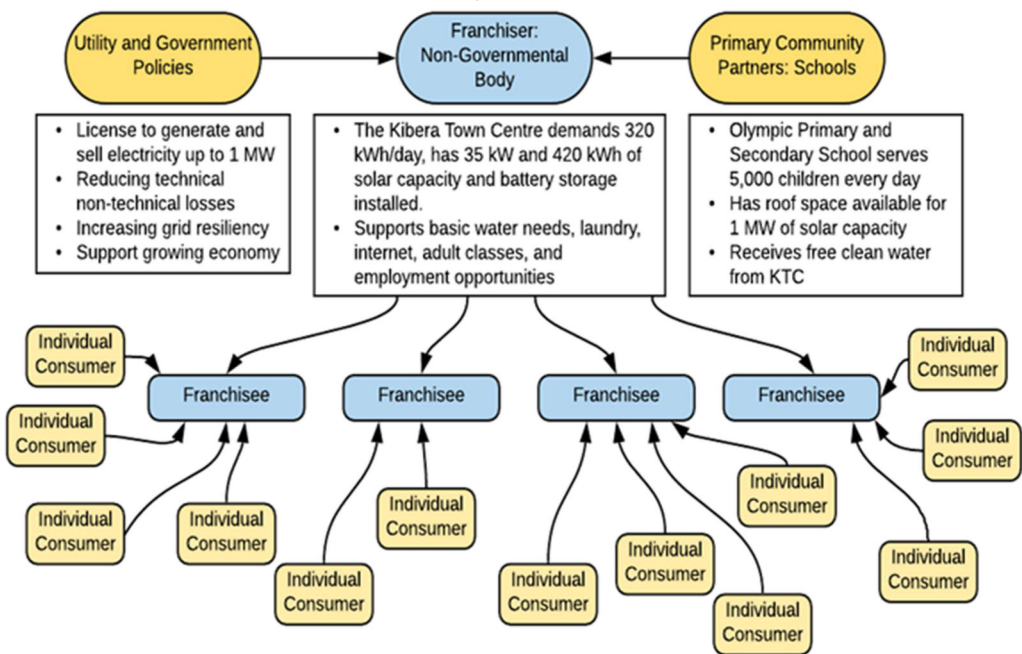
Given electricity loads, resource availability, and costs, the techno-economic simulation results in optimal, least-cost combinations of renewable resources to satisfy demand. However, these simulations do not on their own include aspects such as employment or safety that are

deeply intertwined with energy infrastructure. In informal settlements, the widespread presence of poorly maintained and life-threatening infrastructure limits the appropriateness of raw simulation results. Social and business analyses, in addition to HOMER Pro simulations, are necessary when developing resilient and comprehensive business cases (Bhattacharyya 2015). In Section 6, we use the technical feasibility demonstrated in this section to inform a subsequent opportunity: an electricity franchise business model that serves as a novel, socially resilient solution and public-private partnership for scaling electricity provision in informal settlements.

6. MINIGRID FRANCHISE BUSINESS MODEL

In this research’s key contribution, we propose a new way to approach informal settlement electricity infrastructure in collaboration with communities, NGOs, and government stakeholders through a minigrid franchise business model. Figure 7 conceptualizes our framework for the minigrid franchise business model and the benefits to key stakeholders. Table 5 expands upon the model attributes in detail. Appendix C additionally provides a detailed cost-benefit analysis. In this model, the utility and government will first grant the NGO with a license to generate and sell electricity under the Energy Act 2019. It may be necessary for the NGO to create and spin off a partnered for-profit social enterprise in order to enact this process. The NGO (i.e. the social enterprise) acts as a franchiser and will rely on its established community engagement and trust. Microbusinesses are selected as franchisees that agree to connect and regulate payments for individuals and businesses that opt into the community minigrid. The NGO and other key community loads will be able to sustainably consume high-quality power

FIGURE 7
Framework for a Minigrid Franchise Model at KTC.



A proposed innovative partnership between utilities, NGOs, and the community for the Olympic Neighborhood of Kibera.

to support basic community needs, empowerment services, and education. In doing so, they will reduce non-technical losses, increase grid resiliency, and support a growing local economy.

From stakeholder mapping efforts (See SI), we find that the franchise model's key drivers of success will be the community's need for safe wires, better maintenance of the minigrid equipment, shifting a utility liability into a valuable community asset, and increasing opportunities for skilled employment. KTC's location in one of the most prominent urban areas in East Africa will further facilitate access to materials and training opportunities.

The minigrid simulation and analysis in Sections 5 demonstrates that the addition of the community franchise model is technically and economically feasible. While overall long-term value is difficult to ensure, below we demonstrate why this franchise business model will also be environmentally sustainable, socially and politically resilient, and will support positive health and economic development outcomes.

For every year in operation, *ComCons* has the potential to offset a minimum of 100 kg CO, 620 g particulates, 14 g sulfur, and 95 kg NO_x from reduced diesel consumption as well as 13,000 kg CO₂ from net negative utility consumption over *Ref.*¹¹ In addition to reducing global greenhouse gas emissions, eliminating local pollution reduces community exposure to PM_{2.5}, SO_x, and NO_x, which are known to negatively impact health by increasing the incidence of respiratory diseases and cancers (WHO 2018). These air quality gains are particularly important for a city like Nairobi that suffers from chronically poor air quality. Farquharson et al. (2018) examined the climate impact of the use of diesel generators to back up unreliable grids across Africa. They found that using diesel generators increases fossil fuel consumption compared with the grid by a factor of 1.5–1000 depending on the location. Limiting the use of backup diesel generators with renewable energy will thereby improve local community health and environmental quality.

This community franchise model has additional political and economic benefits. The utility and local government are predisposed to support the model because of the potential for decreasing their liability of working in informal settlements, where both cope with high non-technical losses and maintenance issues. At the same time, by providing agency to slum dwellers, the franchise business model can legalize and legitimize their means of energy access (Cubitt 2013). Creating a clean source of reliable, affordable electricity stimulates local economic activity because businesses will not be subjected to extended periods of time with power outages or poor power quality (Gunning 2014). Community members and microenterprises can benefit from the reliable electricity as well as the potential for a supplemental income from becoming a franchisee or skilled worker for the minigrid. The legal status of the many integrated internally displaced people living in Kibera inhibits their ability to get other skilled jobs (Kamungi 2013), making this opportunity unique and meaningful.

A lack of employment opportunities is a key challenge for lifting Kibera's residents out of poverty. Danner et al. (2016) found that 40% of youth are self-employed in Nairobi's slums. Jobs available to them in public transportation, delivery, solid waste management, food, and construction pay the equivalent of \$3 to \$50 USD daily. In *ComCons*, the additional profit margin from generating and selling excess solar energy is sufficient to support multiple incomes in this range, indicating the potential for supplemental income for franchisees and financial sustainability for KTC.

11. HOMER Pro generator emissions <https://www.homerenergy.com/products/pro/docs/latest/generator.html>.

TABLE 5
Minigrid franchise business model details.

Term	3–7 years, adjustable based on ability to recover upfront investment
Awarding process	Concept and design is tested with broader private sector and wide consultation network Partner is selected based on above feedback and lowest capital cost for components
Upfront investment	KTC, the private partner, carries the investment to expand electricity generation, delivery responsibilities, and other commercial risks. Investment based on existing capacity. Government payments to concessionaires (KTC) through agreed quality of energy services, as outlined by utility. For example, an agreed amount of renewable generation per month and percentage of nontechnical losses. Sources of finance are equity and debt finance, project grants, various external funding.
Operations and maintenance costs	Recipients pay for opting into the community minigrid, local technicians are paid for maintenance and collect tariffs from customers. The utility and the government schools are the main public partners. Schools lease roof space for solar system expansion, and utilities adhere to net metering and local energy generation agreements as legalized in the Energy Act 2019. Utility compensates for proper maintenance of distribution lines while private partner (KTC) implements and takes responsibility for upkeep
Electricity tariff charged	Initially—electricity tariff for customers is equivalent to the cost of existing informal connections to ensure affordability and cost recovery. Long-term—adjustable based on the community value from increased safety and livelihood support from the reliable and maintained service.
Expected electricity served	Estimated from 2018–2020 utility bills, sensor generated daily load profile, extensive energy audits, and generator fuel receipts

The KTC community center is well suited to fill the niche of the franchisor in the community with its strong network, leniency in terms of the legal status of its members, and mission to enhance community welfare. In addition to its other services, the KTC also has a micro-finance department that allows people to borrow money with very minimal qualifications, providing a critical service for many who are turned away from standard banks. Since KTC already provides credit access for this disenfranchised population, they have the capacity and network for extension of credit services through the minigrid franchise model. The most likely early adopters of the minigrid franchise will be the small businesses that rely on electrical appliances to operate as they will be able to quickly actualize the benefits of reliable electricity access.

While adoption of the franchise model is not a given, we are optimistic about its success, in part because it addresses all six of the barriers faced by slum dwellers in Mumbai that limited their access to a regular electricity connection found by Mimmi (2014). Their survey of 3000 slum dwellers in Mumbai confirms that affordability, lack of homeownership status, precarious house construction, pressure from local leaders, and existing social exclusion in governance plays a role in determining an electricity provider. We posit that these barriers and desires are also present in Kibera, and that our proposed solution is uniquely well-suited to overcome these barriers to achieve sustainable electricity access.

7. LIMITATIONS AND OPPORTUNITIES FOR FUTURE WORK

Access to sustainable and reliable electricity (SDG 7) can bolster communities like Kibera, but on its own, electricity is insufficient for supporting a higher quality of life. Rather, electricity access is crucial to powering progress towards many other SDGs; notably education (SDG 4), access to clean water and sanitation (SDG 6), and access to health services (SDG 3) (World Bank 2017). This research shows further an opportunity for renewable energy to additionally support efforts reducing inequality (SDG 10) and making settlements inclusive, safe, and resilient (SDG 11). Future research on sustainable development for under-resourced populations must be approached with an equal awareness of the multitude of intersecting issues faced by communities in informal settlements on a daily basis.

This case study of the operation and potential expansion of KTC's minigrid demonstrates the unique ability of sustainable minigrids to be appropriate solutions for urban environments in informal settlements—technically, economically, and socially. However, transferring the positive impacts found in this research to other locations will depend on the social capital of the partnering organization, resources and costs, and policy. KTC is quite unique in its partnership with, and respect from, the local community of Kibera. Strong partnerships with enduring infrastructure like KTC are crucial for flexible electricity planning that serves characteristically transient populations. The optimal minigrid sizing will depend upon both local renewable energy resources and the local cost of technology. For example, other locations may be better suited to hydropower than solar. Finally, a clear policy landscape such as in Kenya's Energy Act 2019 is needed to create an enabling environment for community minigrids and alternative means of producing and delivering clean electricity.

A limitation of this research surrounds the use of simulation tools such as HOMER Pro. It is impossible to fully capture reality's complexities and nuances in HOMER Pro simulations, particularly with changing costs and load forecasts. Future decisions will require that simulation tools reflect updates to the cost and availability of key components, which vary both temporally and spatially and are influenced by the degree of urbanization. For this particular research, the simulation may need to be modified to account for changes to the tariffs set by the utility, maintenance costs, and the decreasing cost of solar and storage technologies as they arise. The incorporation of life-cycle costs and externalities may better reflect long-term comprehensive impacts to the community. Appendix C introduces a framework for cost-benefit calculations, highlighting the benefits of *ComCons* over the existing infrastructure. Load forecasts may need to be adjusted as more real-time data is collected and services at KTC become more energy efficient or as demand increases. As KTC embarks on expanding their renewable electricity services and the energy policy landscape inevitably shifts, future research will be able to refine the franchise business model in order to analyze the broader opportunity to provide sustainable electricity in informal settlements and quantify the diverse, indirect benefits of urban minigrids such as by providing ancillary service to the electric utility or increasing local safety.

Though many quantitative minigrid studies proffer potential, they are often limited by the scope of model assumptions, empirical data, and the exclusion of key historical, social, and political context. Our mixed-methods approach to minigrid simulation alongside a novel franchise model surmounts these barriers. However, with any project in an informal settlement, additional contextual understandings must be considered such as the interests of individuals and business leaders who, with conditions of informality and weak rule of law, manage (and profit from) the status-quo's insufficient supply of basic services. Other agents including land-

lords are extremely influential in supplier choice, adding to the limited decision-making power of households in informal settlements. Understanding past failures involves an historical and critical perspective of other slum ‘improvement’ projects, which have left a legacy of skepticism (Mimmi 2014).

Finally, more analysis is needed to validate the minigrid franchise model and expansion plan’s differential benefits across genders. Clean, reliable electricity can create safer societies for women and girls by increasing female access to opportunities, education and resources, and strengthening their participation in policy- and decision-making (Almansi et al. 2020; Saini et al. 2019). In 2019, KTC installed local solar street lighting to promote safety particularly for women. KTC’s clean water and sanitation services and employment opportunities continue to work towards gender equality. Building from this article, future research is needed to examine the dynamics between gender and energy in informal settlements in greater depth.

This research demonstrates the use of interdisciplinary methods to inform the economic and technical feasibility of an innovative, financially and environmentally sustainable minigrid franchise model that empowers and employs those living in urban informal settlements. This minigrid franchise model is best suited to serving slum communities and refugee camps around the world that have similar sustainable city goals, infrastructure, legal concerns, NGO presence for basic utilities and key community loads, as well as private-public partnerships. In order to fully achieve sustainable cities, we must account for the enduring presence of informal settlements, which need thoughtful approaches and directed solutions. Further examination of the structures providing affordable clean water and sanitation services along with renewable, reliable electricity access to informal settlements can reveal effective, appropriate solutions for developing healthy, resilient communities.

✎ 8. CONCLUSION ✎

This research details the extensive monitoring and evaluation efforts that can inform future resilient and sustainable urban planning for informal settlements. We manifest the benefits of investing in collaborations and ethically gathering electricity data, which can serve as a model for future city planners and basic needs providers. The simulation and optimization of the KTC energy system encourages its expansion to serve the Olympic Primary and Secondary Schools and 50 local businesses. NPV calculations indicate that KTC could save over \$38,000 USD over the next 25 years by implementing the community minigrid instead of maintaining the status-quo system. Opportunity for future expansion to balance load growth lies in the remaining 60% of Olympic Primary School’s roof space and the entirety of Olympic Secondary School’s roof space. By leasing this extra roof to other small businesses or other organizations that lack space for solar panels, KTC can stimulate growth of more private-public partnerships, and further increase the positive socio-economic impact of the minigrid and community.

While we cannot change the legal status of communities in informal settlements to bring them more rights and political decision-making power, we can include them in plans to extend basic electricity services in a way that decriminalizes their means of access. Additionally, through the Energy Act 2019, electricity access can be more affordable, reliable, sustainable, and modern than the status-quo. Environmentally sustainable and socially resilient planning for urban informal settlements hinges on enabling policies combined with applied research and informed business models to leverage opportunities for equitable transitions toward sustainable urban development.

Sustainable cities and affordable, reliable, sustainable, and modern energy access for all will rely on inclusive, scalable generation and distribution models for productive power to support safe, healthy livelihoods and resilient communities.

✎ SUPPLEMENTARY INFORMATION (SI) ✎

Can be found ADD LINK HERE!.

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Both publications have earned SCIMago Journal Ratings in the top quartile for Economics and Econometrics publications.

IAEE wishes to congratulate and thank all those involved including authors, editors, peer-reviewers, the editorial boards of both publications, and to you, our readers and researchers, for your invaluable contributions in making 2019 a strong year. We count on your continued support and future submission of papers to these leading publications.